The Grok Portfolio White Paper

The Grok Portfolio follows an investment strategy that harnesses Grok's analytical capabilities. By leveraging the latest advancements in artificial intelligence, the system provides systematically curated stock and sector selections. One of its standout features is using LLMs to derive investment scores that guide portfolio decisions.

Core Investment Rules

The portfolio uses a structured investment approach:

- Composition: 15 individual assets that can include individual stocks or ETFs selected by Grok per the analysis process discussed below.
- Holding Period: All assets are held for one month
- Monthly Rebalancing: At month-end, the system conducts a complete analysis to select the top 15 assets and rebalances accordingly

AI-Powered Analysis System

A key distinction of our system is its method of keeping Grok current with market conditions. Since these models don't have real-time market knowledge, we provide them with the latest information through a systematic process:

The analysis integrates multiple data streams:

1. Economic Context
    - Comprehensive AI-generated economic reports using the latest information about current market conditions and global events
2. Company-Level Information
    - Financial ratios and metrics
    - Recent price movements and technical indicators
    - Current news and announcements

Scoring Methodology

The system employs carefully crafted prompts to ensure consistent and meaningful analysis. The prompts used to generate analyses of certain sectors and individual companies are disclosed in Exhibit 1.

Portfolio Construction Process

The system follows a systematic monthly process:

1. Macro Analysis
    - Collection of economic indicators and trends
    - Analysis of significant global events
    - Generation of a comprehensive economic report
2. Individual Stock Score
    - Analysis of S&P 500 companies
    - Integration of company-specific news and data
    - AI scores using firm-specific, sector, and macro information
    - Pre-selection of the top 30 companies based on the scores
3. Portfolio Balance
    - Allocation is decided by LLMs to provide a diversified portfolio that does well under current conditions
    - Monthly rebalancing to maintain target allocations

Risk Management

The system incorporates several risk management features:

- Diversification across multiple sectors and companies
- Regular monthly rebalancing
- Systematic scoring process
- Multiple data sources

Conclusions

The Grok Portfolio represents a shift in investment management, placing AI at the core of the investment process. Through its analysis and the unparalleled capabilities of LLMs, it offers a systematic approach to navigating market complexities. While all investing carries inherent risks, the Grok Portfolio leverages cutting-edge technology to provide a disciplined, data-driven investment approach.

The system's success relies on three key elements:

1. Systematic data collection and processing
2. Sophisticated AI analysis through structured prompts
3. Disciplined portfolio construction and rebalancing

This combination of technological innovation and investment discipline creates a unique approach to portfolio management that aims to capitalize on the analytical power of AI while maintaining rigorous investment principles.

Exhibit 1 - Prompts

Firm prompt:

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      "Pretend you are a financial expert with stock recommendation experience.\n"

      "Speak in the third person."
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        "You do not mention your credentials.\n"

        f"Macro-economic data for context:\n{macro_news}\n"

        "Based on the provided financial data and news headlines,"

        " please assign a score (from 1 to 100) reflecting the potential
investment value of"

        f" company {company_name} in the {industry} industry for the next
month.\n"

        "First, write a short investment report about the firm
situation.\n"

        "Include sections of recent news, financials, valuations, and
economic outlook affecting the firm.\n"

        "Do not recommend alternatives.\n"

        "Do not mention the word 'provided' instead use 'recent' or
'latest'."

        "Do not speak directly to investors nor recommend actions.\n"

        "Start with 'Investment Report:'\n"

        "Finally, in a new line, output Score: X."
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Exhibit 2 - Data sources

1. Wikipedia
2. https://stocknewsapi.com/
3. https://finance.yahoo.com/

Exhibit 2A

For each firm, the news inputs are all past week's news headlines with summaries from the following news outlets:

1. 24/7 Wall Street
2. Accesswire
3. After Earnings (video)
4. Barrons
5. Benzinga
6. Bloomberg Markets and Finance (video)
7. Bloomberg Technology (video)
8. Business Insider
9. Business Wire
10. CNBC
11. CNBC International TV
12. CNBC Television
13. CNET
14. CNN
15. CNN Business
16. Cheddar Videos
17. Deadline
18. Digital Trends (video)
19. Discount The Obvious
20. ETF Trends
21. ETF.com
22. Engadget (video)
23. Fast Company
24. Finbold
25. Forbes
26. Fox Business
27. FreightWaves
28. FXEmpire
29. GeekWire

30. Globe News Wire
31. Green Stock News (video)
32. GuruFocus
33. Huffington Post
34. InsiderTrades
35. Investopedia
36. Investor Place
37. Investors Business Daily
38. Invezz
39. Kiplinger
40. Kitco
41. Marijuana Stocks
42. Market Watch
43. Mcap MediaWire
44. Millennial Money
45. Morningstar Inc.
46. New York Times
47. New York Post
48. Newsfile Corp
49. MarketBeat
50. MCAP MediaWire
51. Penny Stocks
52. Proactive Investors
53. PYMNTS.com
54. PR Newswire
55. Pulse2
56. Reuters
57. Schaeffers Research
58. Schwab Network (video)
59. See It Market
60. Seeking Alpha
61. Skynews
62. Stock Market.com
63. TechCrunch
64. TechXplore

65. The Dog of Wall Street
66. The Financial News
67. The Guardian
68. The Motley Fool
69. The Street (video)
70. The Verge (video)
71. VentureBeat
72. Wall Street Journal (articles)
73. Wall Street Journal (video)
74. Yahoo Finance (video)
75. Zacks Investment Research

Exhibit 2B

For firms financials, we use the most recent available information about these variables:

1. Previous Close Price
2. Opening Price
3. Lowest Price of the Day
4. Highest Price of the Day
5. Previous Close Price (Regular Market)
6. Opening Price (Regular Market)
7. Lowest Price of the Day (Regular Market)
8. Highest Price of the Day (Regular Market)
9. Annual Dividend Rate
10. Dividend Yield
11. Payout Ratio (Dividend Payout as Percentage of Earnings)
12. Five-Year Average Dividend Yield
13. Beta (Volatility Measure)
14. Price-to-Earnings Ratio (Trailing 12 Months)
15. Price-to-Earnings Ratio (Forward 12 Months)
16. Daily Volume of Shares Traded
17. Daily Volume of Shares Traded (Regular Market)

18. Average Volume of Shares Traded
19. Average Volume of Shares Traded (Last 10 Days)
20. Average Daily Volume of Shares Traded (Last 10 Days)
21. Bid Price
22. Ask Price
23. Bid Size (Number of Shares)
24. Ask Size (Number of Shares)
25. Market Capitalization
26. 52-Week Low Price
27. 52-Week High Price
28. Price-to-Sales Ratio (Trailing 12 Months)
29. 50-Day Moving Average Price
30. 200-Day Moving Average Price
31. Annual Dividend Rate (Trailing 12 Months)
32. Dividend Yield (Trailing 12 Months)
33. Trading Currency
34. Enterprise Value
35. Profit Margins
36. Float Shares (Shares Available to Public)
37. Shares Outstanding
38. Shares Sold Short
39. Shares Sold Short in the Previous Month
40. Short Interest Date of Previous Month
41. Most Recent Short Interest Date
42. Short Interest as Percentage of Outstanding Shares
43. Insider Holdings Percentage
44. Institutional Holdings Percentage
45. Short Ratio (Days to Cover Short Positions)
46. Short Interest as Percentage of Float
47. Implied Shares Outstanding
48. Book Value per Share
49. Price-to-Book Ratio
50. Last Fiscal Year End Date
51. Next Fiscal Year End Date
52. Most Recent Quarter Date

53. Earnings Growth (Quarterly)
54. Net Income to Common Stockholders
55. Earnings per Share (Trailing 12 Months)
56. Earnings per Share (Forward 12 Months)
57. Price/Earnings to Growth (PEG) Ratio
58. Last Stock Split Factor
59. Last Stock Split Date
60. Enterprise Value to Revenue Ratio
61. Enterprise Value to EBITDA Ratio
62. 52-Week Price Change Percentage
63. S&P 500 52-Week Price Change Percentage
64. Last Dividend Value
65. Last Dividend Payment Date
66. Highest Target Price (Analyst Forecast)
67. Lowest Target Price (Analyst Forecast)
68. Mean Target Price (Analyst Forecast)
69. Median Target Price (Analyst Forecast)
70. Recommendation Mean (Analyst Consensus)
71. Recommendation Key (Analyst Rating)
72. Number of Analyst Opinions
73. Total Cash
74. Cash per Share
75. EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization)
76. Total Debt
77. Quick Ratio (Liquidity Measure)
78. Current Ratio (Liquidity Measure)
79. Total Revenue
80. Debt-to-Equity Ratio
81. Revenue per Share
82. Return on Assets (ROA)
83. Return on Equity (ROE)
84. Free Cash Flow
85. Operating Cash Flow
86. Earnings Growth Rate
87. Revenue Growth Rate

88. Gross Profit Margins
89. EBITDA Margins
90. Operating Margins
91. Financial Currency
92. Price/Earnings to Growth (PEG) Ratio (Trailing)
93. Audit Risk Level
94. Board Risk Level
95. Compensation Risk Level
96. Shareholder Rights Risk Level
97. Overall Risk Level

Exhibit 2C

For the macro news, we use the past week's news about general markets from the same sources as in Exhibit 2A. Additionally, we include current events from
https://en.wikipedia.org/wiki/2025,
https://en.wikipedia.org/wiki/2024_in_the_United_States

https://en.wikipedia.org/wiki/2025,
https://en.wikipedia.org/wiki/2025_in_the_United_States

Exhibit 2D

For the macroeconomic events, we additionally use DeeperSearch from grok-3 with the following prompt:

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Here are some events and context to update your knowledge information
cutoff.

(Insert info here from previous steps)

Provide a complete expected timeline of the most important economic and
political events for the next three months in the USA. Not only the
scheduled events and known forecasts. Also, provide your best expectations
about the realization of these events. Pay special attention to the next
month. I also want a table with your forecast for interest rates,
inflation, tariffs, and other economic events for the next month and
quarter. Not only what analysts and the market expect. Provide your
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expectations based on your research and compare them with the market's expectations.

Exhibit 2E

We then use the following prompt for the final allocation

Now, I want a 15-asset portfolio where we will invest for the next month (rebalancing in one month) in a table with weight, thesis, edge, and risk. Weight this portfolio to perform positively given the market conditions and to beat the S&P 500. We have the following reports for the stocks that were scored the highest:

(Insert top scores here)

However, we can also invest in most ETFs (except short, leveraged, or vol because of the monthly horizon), including but not limited to market, sectors, TIPS, and long and short-term bonds. You decide the weights do not have to include any or all of the stocks or instruments mentioned.

Remember, a 15-asset portfolio where we will invest for the next month (rebalancing in one month) in a table with weight, instrument type, thesis, edge, and risk. You can use both the stock info and your macro expectations, as well as reasoning about the future and markets.

Exhibit 4

History of trades can be found on the Autopilot app here.